

Data dictionary

Client: One row represents one client (a buyer or seller) registered with TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Client_id	Unique identifier	Integer	PK	NOT NULL	1001
First_name	First name of client	Varchar		NOT NULL	John
Last_name	Last name of client	Varchar		NOT NULL	Doe
email	Contact email	Varchar		NOT NULL	Johndoe@gmail.com
phone	Contact email	varchar		NOT NULL	403-123-4567
Created_at	When client account was created	date		NOT NULL	2026-10-06

Agent : One row represents an independent real estate agent affiliated with TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Agent_id	Unique identifier	Integer	PK	NOT NULL	1001
Licence number	Agents' real estate licence	Varchar		NOT NULL	RE-ag-123
First_name	first name of agent	integer		NOT NULL	Agent
Last_name	Last name of agent	Varchar		NOT NULL	bob
Email	Contact email	varchar		NOT NULL	Agentbob@gmail.com
Phone	Contact phone	varchar		NOT NULL	403-123-4567
Agent_status	Is the agent active	varchar		NOT NULL	Active

Staff : One row represents an internal, full-time administrative or operations employee at TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Staff_id	Unique identifier	Integer	PK	NOT NULL	1001
First_name	Staffs first name	Varchar		NOT NULL	Staff
Last_name	Staffs last name	Varchar		NOT NULL	member
email	Contact email	Varchar		NOT NULL	Staffmember@gmail.com
Role_code	Unique number that specifies role	Varchar		NOT NULL	EMP-123
Is_active	Current employment status	Varchar		NOT NULL	Active

Property : One row represents a unique real estate asset managed in the system

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Property_id	Unique identifier	integer	PK	NOT NULL	1001
Adress_text	Property street address	Varchar		NOT NULL	67 tickle boulevard
Property_ref	Municipal property code	Varchar	UK	NOT NULL	A-123
Municipality	City where property is located	Varchar		NOT NULL	Calgary
Property_type	Tupe of property(apartment, house etc)	Varchar		NOT NULL	Condo
bedrooms	Amount of bedrooms	numeric		Must be greater than or equal to zero	2
bathrooms	Amount of bathrooms	numeric		Must be greater than or equal to zero	2
Floor_area_sqft	Total square feet in property	numeric		Must be > zero	1000sqft

Listing : One row represents an individual active market listing for a property

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Listing_id	Unique identifier	integer	PK	NOT NULL	1001
Property_id	Foreign key referencing the property	integer	FK	NOT NULL	1234
Agent_id	Foreign key referencing the agent	integer	FK	NOT NULL	6769
Mls_number	Mls registration number	varchar		UNIQUE	A01234
Asking_price	Initial or current listed sale price	numeric		NOT NULL	650,000
Listed_on	Date the listing went on the market	date		NOT NULL	2026-10-06
Closed_on	Date the listing was sold or withdrawn	date		Optional	2026-11-06
Listing_status	Status of listing	varchar		NOT NULL	Active

Listing seller: One row represents an associative link attaching one seller client as an owner of a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Listing_id	Foreign key identifying listing	integer	FK	NOT NULL	1001
Client_id	Foreign key identifying client	integer	FK	NOT NULL	1234

Showing : One row represents a scheduled viewing appointment time slot hosted by an agent for a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Showing_id	Unique identifier	Integer	PK	NOT NULL	1001
Listing_id	Foreign key identifying listing	Integer	FK	NOT NULL	1234
Agent_id	Foreign key identifying agent	Integer	FK	NOT NULL	6769
Starts_at	Date and time the showing begins	timestampz		NOT NULL	2026-10-6
Ends_at	Date and time the showing ends	timestampz		NOT NULL	2026-11-6
Showing_status	Outcome of showing	varchar		NOT NULL	active showing

Showing Booking: One row represents a client's specific booking for a property showing appointment

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Booking_id	Unique identifier	integer	PK	NOT NULL	1001
Showing_id	Foreign key indentifying showing	Integer	FK	NOT NULL	1234
Client_id	Foreign key identifying client	Integer	FK	NOT NULL	6769
Party_size	Amount of people viewing property	numeric		NOT NULL	5
Showing_status	Outcome of showing	varchar		NOT NULL	active showing
Booked_at	Time the showing is booked at	timestampz		NOT NULL	2026-10-10

Offer : One row represents a formal purchase proposal submitted by an agent for a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Offer_id	Unique identifier	integer	Pk	NOT NULL	1001
Listing_id	Foreign key identifying listing	integer	Fk	NOT NULL	1234
Agent_id	Foreign key identifying agent	integer	FK	NOT NULL	6768
Offer_amount	Proposed purchase amount	numeric		NOT NULL	650,000
Submitted_at	Timestamp when offer was sent	timestampz		NOT NULL	2026-10-06
Expires_at	Timestamp when offer expires	timestampz		Optional	2026-11-06
Offer_status	Current processing status of the offer	Varchar		NOT NULL	Active
Accepted_at	Timestamp when the offer was accepted	timestampz		Optional	2026-10-21

Offer buyer : One row represents an associative link attaching one buyer client to a submitted purchase offer

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Offer_id	Foreign key identifying offer	integer	FK	NOT NULL	1001
Client_id	foreign key identifying client	integer	FK	NOT NULL	1234

Sale transaction :One row represents a completed, real estate sale finalized from an accepted purchase offer

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
Sale_id	Unique identifier	Integer	PK	NOT NULL	1001
Offer_id	Foreign key identifying offer	integer	FK	NOT NULL	1234
Coordinator_staff_id	Foreign key identifying staff	integer	FK	NOT NULL	6769
Completed_on	official date of transaction	date		NOT NULL	2026-10-06
Sale_price	Final price of the sale	numeric		NOT NULL	650,000